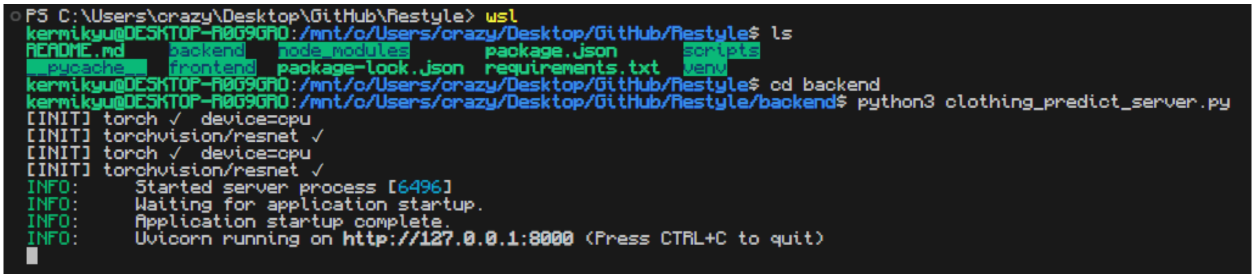
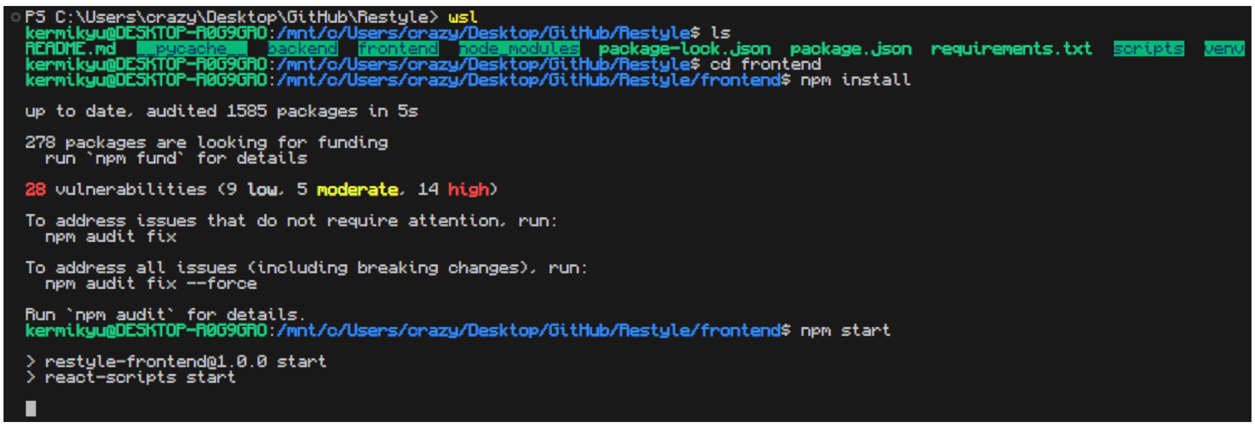
## Activating the Python file that scans the clothes



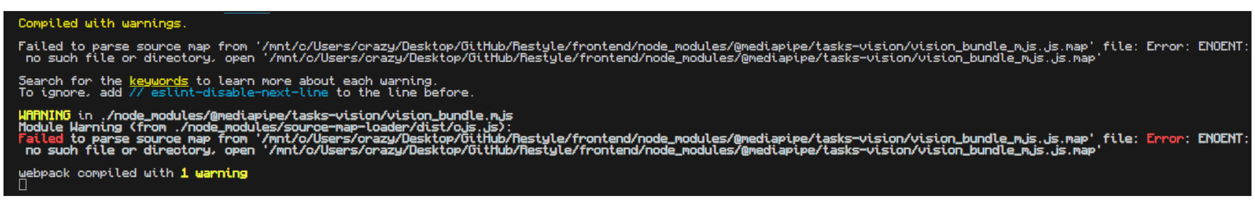
```
PS C:\Users\crazy\Desktop\GitHub\Restyle> wsl
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ ls
README.md     backend    node_modules       package.json       scripts
__pycache__   frontend   package-lock.json  requirements.txt   venv
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ cd backend
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle/backend$ python3 clothing_predict_server.py
[INIT] torch ✓  device=cpu
[INIT] torchvision/resnet ✓
[INIT] torch ✓  device=cpu
[INIT] torchvision/resnet ✓
INFO:     Started server process [6496]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

## Starting the frontend



```
PS C:\Users\crazy\Desktop\GitHub\Restyle> wsl
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ ls
README.md  __pycache__  backend  frontend  node_modules  package-lock.json  package.json  requirements.txt  scripts  venv
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ cd frontend
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend$ npm install

up to date, audited 1585 packages in 5s

278 packages are looking for funding
  run `npm fund` for details

28 vulnerabilities (9 low, 5 moderate, 14 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
kermikyu@DESKTOP-A0G9GAO:/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend$ npm start

> restyle-frontend@1.0.0 start
> react-scripts start
```

## Warnings, however, the website does work as intended



```
Compiled with warnings.

Failed to parse source map from '/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend/node_modules/@mediapipe/tasks-vision/vision_bundle_mjs.js.map' file: Error: ENOENT:
 no such file or directory, open '/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend/node_modules/@mediapipe/tasks-vision/vision_bundle_mjs.js.map'

Search for the keywords to learn more about each warning.
To ignore, add // eslint-disable-next-line to the line before.

WARNING in ./node_modules/@mediapipe/tasks-vision/vision_bundle.mjs
Module Warning (from ./node_modules/source-map-loader/dist/cjs.js):
Failed to parse source map from '/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend/node_modules/@mediapipe/tasks-vision/vision_bundle_mjs.js.map' file: Error: ENOENT:
 no such file or directory, open '/mnt/c/Users/crazy/Desktop/GitHub/Restyle/frontend/node_modules/@mediapipe/tasks-vision/vision_bundle_mjs.js.map'

webpack compiled with 1 warning
```

## Started and connected to the backend

```
PS C:\Users\crazy\Desktop\GitHub\Restyle> wsl
kermikyu@DESKTOP-A0G9GA0:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ ls
README.md  __pycache__  backend  frontend  node_modules  package-lock.json  package.json  requirements.txt  scripts  venv
kermikyu@DESKTOP-A0G9GA0:/mnt/c/Users/crazy/Desktop/GitHub/Restyle$ cd backend
kermikyu@DESKTOP-A0G9GA0:/mnt/c/Users/crazy/Desktop/GitHub/Restyle/backend$ npm install

up to date, audited 204 packages in 1s

33 packages are looking for funding
  run `npm fund` for details

1 moderate severity vulnerability

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
kermikyu@DESKTOP-A0G9GA0:/mnt/c/Users/crazy/Desktop/GitHub/Restyle/backend$ npm start

> restyle-backend@1.0.0 start
> node server.js


✓ Restyle Backend running at http://localhost:5000

Connected to PostgreSQL database
Database tables initialized
```

# README FILE

````
# Restyle - Wardrobe Manager

A simple, clean wardrobe management application built with React and Node.js. Manage your clothing inventory, analyze outfit compatibility, and get capsule wardrobe recommendations.

## Features

- **Wardrobe Inventory** - Track clothing items by category, color, and season
- **Outfit Analysis** - See how items work together
- **AI Outfit Generation** - Use beam search to create outfit combinations
- **Capsule Recommendations** - Get suggestions for a minimal wardrobe
- **Image Upload** - Add photos of your clothing items

## Tech Stack

**Frontend:** React 18, React Router, Axios, CSS3
**Backend:** Node.js, Express, PostgreSQL, Sharp (image processing)

## Quick Start

### 1. Setup Database
The app includes sample data for testing. When you start the backend, it will automatically create tables and insert sample clothing items.

### 2. Start the Application

**Terminal 1 - Backend:**
```bash
cd backend
npm install
npm run dev
```

**Terminal 2 - Frontend:**
```bash
cd frontend
npm install
npm start
```


### 3. Test the Outfit Generator
- Go to the Inventory page
- You'll see 10 sample clothing items
- Click "Generate Outfits" to see AI-generated combinations

````

# README FILE PT.2

````markdown
## Prerequisites

- Node.js v14+
- PostgreSQL
- npm or yarn

## Quick Start

### 1. Clone the Repository

```bash
git clone https://github.com/yourusername/project-website.git
cd project-website
```

### 2. Setup Environment Variables

```bash
cp .env.example .env
```

Edit `.env` with your configuration values.

### 3. Install Frontend Dependencies

```bash
cd frontend
npm install
```

### 4. Install Backend Dependencies

```bash
cd ../backend
npm install
```

````

# README FILE PT.3

````markdown
### 5. Setup Python Environment (prediction service)

The clothing prediction service is a FastAPI app located at `backend/clothing_predict_server.py`.

```bash
# from the repo root
cd backend
# optional: create and activate a virtual environment
python3 -m venv venv
source venv/bin/activate    # macOS / Linux
# venv\Scripts\activate    # Windows (PowerShell)

pip install -r requirements.txt

# Optional: install CPU PyTorch for ImageNet model (no GPU/triton required)
pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu

# Start the predictor (FastAPI on port 8000)
python3 clothing_predict_server.py
```

## Running the Application (local)

Run each block in its own terminal.

### Terminal 1 — Frontend (Port 3000)
```bash
cd frontend
npm install
npm start
```

### Terminal 2 — Backend Node API (Port 5000)
```bash
cd backend
npm install
npm run dev
# server.js listens on PORT (default 5000)
```

````

# README FILE PT.4

~~~~markdown
### Terminal 3 — Python predictor (FastAPI, Port 8000)
```bash
cd backend
source venv/bin/activate   # if you created one earlier
python3 clothing_predict_server.py
```

The frontend is available at `http://localhost:3000` and the backend API at `http://localhost:5000`.
The prediction service runs at `http://127.0.0.1:8000/predict-clothing` by default.

Quick health/debug endpoints:

```bash
# Node backend health
curl http://localhost:5000/api/health

# Python predictor debug/status
curl http://127.0.0.1:8000/debug/status
```

Troubleshooting notes:
- If CLIP fails to import with a Triton error (`No module named 'triton.backends'`), uninstall Triton in your Python environment:

```bash
pip3 uninstall triton -y
```

- If you only need basic ML without CLIP, installing CPU PyTorch (above) gives ImageNet-based predictions without GPU/triton complexities.

## API Endpoints

### Backend (Node.js)

**Clothing Items:**
- `GET /api/clothing` - Get all clothing items
- `POST /api/clothing` - Add new clothing item
- `GET /api/clothing/:id` - Get specific item details
- `DELETE /api/clothing/:id` - Remove clothing item

**Outfit Analysis:**
- `POST /api/analyze/compatibility` - Get outfit compatibility analysis
- `GET /api/analyze/utilization` - Get item utilization metrics

~~~~

# README FILE PT.5

~~~~markdown
**Capsule Wardrobe:**
- `POST /api/capsule/recommendations` - Get capsule wardrobe suggestions
- `GET /api/capsule/:id` - Get generated capsule details

**System:**
- `GET /api/health` - Health check endpoint

## Environment Variables

Create a `.env` file in the root directory:

```env
# Frontend
REACT_APP_API_URL=http://localhost:5000

# Backend
PORT=5000
NODE_ENV=development
DB_USER=postgres
DB_PASSWORD=your_password
DB_HOST=localhost
DB_PORT=5432
DB_NAME=wardrobe_db
CORS_ORIGIN=http://localhost:3000

# Python Engine (IF NEEDED)
OPENAI_API_KEY=your_openai_api_key
PYTHON_ENGINE_PORT=5001
PYTORCH_MODEL_PATH=./models
```

## Database Setup

### PostgreSQL

1. Create a new database:
```bash
psql -U postgres -c "CREATE DATABASE wardrobe_db;"
```

2. Update `.env` with your PostgreSQL credentials

3. The backend will initialize the required tables on first run
~~~~

**README FILE PT.6**

````
### Database Tables
- `clothing_items` - Stores user clothing inventory
- `outfit_compatibility` - Stores outfit combination rules
- `capsule_recommendations` - Stores generated capsule wardrobes
- `user_preferences` - Stores user constraints and preferences

## Deployment

### Deploy to Vercel

1. Connect your GitHub repository to Vercel
2. Set environment variables in Vercel dashboard
3. Deploy frontend and backend separately

**Frontend:**
```bash
cd frontend
npm run build
vercel deploy
```

**Backend:**
```bash
cd backend
vercel deploy
```

## Contributing

Please see [CONTRIBUTING.md](.github/CONTRIBUTING.md) for guidelines on how to contribute to this project.

## License

This project is licensed under the MIT License - see the LICENSE file for details.

## Support

For issues, questions, or suggestions, please open an issue on GitHub.

## Useful Links

````

**README FILE PT.7**

```
## Useful Links

- [React Documentation](https://react.dev)
- [Node.js Documentation](https://nodejs.org/docs)
- [PostgreSQL Documentation](https://www.postgresql.org/docs)
- [PyTorch Documentation](https://pytorch.org/docs)
- [Vercel Documentation](https://vercel.com/docs)

---
```